Unit XVIII: Lipids

Lecture notes: the slides plus what she explained in class (transcript) and wrote on the board

CHEMISTRY (PROF. L. RANAY)

WHAT LIPIDS ARE

- **Lipid** (Greek *lipos*, fat): an organic compound found in living organisms that is **insoluble (or only sparingly soluble) in water** but **soluble in nonpolar organic solvents** (the hydrocarbon part makes them insoluble).

Slide exercise

A lipid is insoluble in water and soluble in nonpolar organic solvents (**d**).

- Unlike the other biomolecules, lipids have **no common structural feature** (no building block), so they **do not form polymers or macromolecules**. (Proteins → amino acids, nucleic acids → nucleotides, carbohydrates → sugars.)
- Lipids are classified in **two ways**: by **biochemical function** and by **structure**.

CLASSIFICATION BY BIOLOGICAL FUNCTION (FIVE)

- **1. Energy-storage lipids: triacylglycerols (triglycerides). Long-term** energy storage (carbohydrates store energy too, but short-term); used during **starvation**. The energy is in the **fatty acid** hydrocarbon chains (broken down by β-oxidation, the spiral pathway).
- **2. Membrane lipids: phospholipids, sphingoglycolipids and cholesterol.** Structural components of cell membranes (the protective barrier around the cell); also part of the outer coating of bacteria, plants, insects and vertebrates.
- **3. Emulsification lipids: bile acids** (and bile salts). **Bile** is a fluid **made in the liver**, **stored in the gallbladder** and **released into the small intestine** during digestion; it breaks large lipid globules into smaller ones so they can be absorbed.
- **4. Messenger lipids: steroid hormones and eicosanoids** (intra- and intercellular signaling).
  - Steroid hormones: **sex hormones** (testosterone, estrogen, progesterone/progestins, androgens) and **adrenocorticoid hormones** (e.g., glucocorticoids).
  - **Eicosanoids** (*eicosa* = **20 carbons**): **prostaglandins** (released in injury, illness, **inflammation**), **thromboxanes** (**blood clotting**, platelet aggregation), **leukotrienes** (**allergy**, immune response, **asthma**, rhinitis).
- **5. Protective coating: biological waxes.** Water-repellent outer coating: **beeswax**, the skin of **apples**, **cabbage** leaves.

THE CELL MEMBRANE (LIPID BILAYER)

- Made of phospholipids. A **glycerophospholipid** has a **glycerol** backbone (an alcohol), a **phosphate** group and **two fatty acids**.
- **Polar head** = the phosphate part (hydrophilic, attracted to water). **Nonpolar tails** = the fatty acids (hydrophobic hydrocarbon chains).
- In the **bilayer**, the polar heads face the water on **both** sides (outside the cell and the cytoplasm inside); the nonpolar tails point **inward**, toward each other.
- **Cholesterol** in the membrane regulates **membrane fluidity**. Factors that affect fluidity: **cholesterol**, **unsaturated fatty acids** and **temperature**. (Cholesterol is not only "bad".)

SPHINGOSINE AND THE SPHINGOLIPIDS

- **Sphingosine** = the backbone of sphingolipids: an **18-carbon, monounsaturated** (one C=C) **aminodiol** (one **amino** group, **two** OH groups).
- **Sphingophospholipid**: sphingosine + fatty acid + phosphate (+ choline). Example: **sphingomyelin**, in the **myelin sheath** of neurons.
- **Sphingoglycolipid**: sphingosine + fatty acid + a **sugar** (no phosphate; "glyco" = sugar). Example: **cerebroside** (with glucose), in the **brain and nervous system**.
- The fatty acid joins sphingosine's NH by an **amide linkage** (the only new linkage; all the others are

ester linkages).

CLASSIFICATION BY STRUCTURE (THREE TYPES)

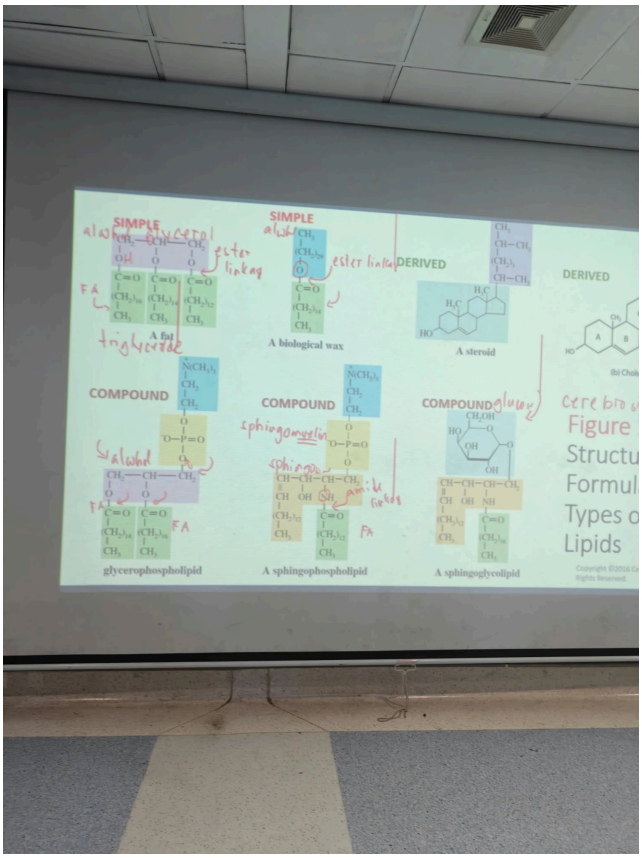
- **Simple lipids: alcohol + fatty acid(s)** joined by **ester linkages**. Examples: **fats (triacylglycerols)** and **biological waxes**.
- **Complex (compound) lipids:** contain **additional groupings**: phosphoric acid, sugars, nitrogenous bases or proteins. Examples: **phospholipids** (glycerophospholipids, sphingophospholipids), **glycolipids** (sphingoglycolipids), **lipoproteins**.
- **Derived lipids:** usually **no fatty acids**: straight hydrocarbons or **fused ring systems**. Examples: **steroids** (cholesterol, sex and adrenocortical hormones), **prostaglandins, thromboxanes, leukotrienes**, and **terpenes** (hydrocarbons with C=C; e.g., **limonene** in lemons and citrus). Steroids count as lipids because they occur in natural fats and share their solubility.
- **Steroid nucleus: three six-membered rings and one five-membered ring**, fused.

LIPOPROTEINS (COMPLEX LIPIDS)

- **Transport** cholesterol (and other lipids) in the blood. **HDL = good** cholesterol; **LDL = bad** cholesterol.
- In a **lipid profile**, high LDL means excess lipid stays in the blood instead of going to the liver → **atherosclerosis** (cardiovascular disease).

READING FIGURE 19.1 (HER BOARD WORK)

- **A fat (triglyceride, simple): glycerol** (alcohol, purple) + **three fatty acids** (green), each by an **ester linkage**.
- **A biological wax (simple):** a long-chain alcohol with **one** OH + **one** fatty acid, by an **ester linkage**.
- **Glycerophospholipid (compound):** glycerol + **two** fatty acids (ester) + **phosphate** + an amino alcohol (choline, $N(CH_3)_3^+$).
- **Sphingophospholipid (compound): sphingosine** + fatty acid (**amide linkage**) + phosphate + choline = **sphingomyelin**.
- **Sphingoglycolipid (compound):** sphingosine + fatty acid (amide) + **glucose** = **cerebroside**.
- **Steroid and cholesterol (derived):** no fatty acid; fused rings.



Her board work on Figure 19.1: alcohol/glycerol, ester linkage, FA, triglyceride, sphingomyelin, sphingosine, amide linkage, glucose, cerebroside

FATTY ACIDS

- A **long hydrocarbon chain** with a **carboxyl group (–COOH) at one end**; building blocks of fats and other lipids; used for **energy** and as **membrane components**.
- Naturally occurring fatty acids are **monocarboxylic**, nearly always have an **even** number of carbons (12, 14, 16, 18…), and are **unbranched** (slide exercise, answer **c**).
- **Amphipathic:** the **carboxyl group is hydrophilic** (polar head), the **hydrocarbon tail is hydrophobic** (nonpolar tail). The carboxyl is a small part, so fatty acids are mostly hydrophobic and insoluble in water.
- **Numbering starts from the –COOH end.** Structures can be drawn extended, condensed or skeletal (palmitic acid on the slide).

SATURATED AND UNSATURATED

- **Saturated:** all C–C bonds in the chain are **single**.
- **Unsaturated: at least one C=C**. **Monounsaturated** = one C=C; **polyunsaturated** = two or more.

Slide exercise

Fatty acids are classified as **saturated, monounsaturated or polyunsaturated** (answer **a**).

- **Omega-3:** the endmost double bond is **3 carbons from the methyl end**. **Omega-6: 6 carbons** from the methyl end.
- Her method: find the **CH_3 at the end of the nonpolar tail**, count from it (C1, C2, C3…): first C=C at the 3rd carbon = omega-3; at the 6th = omega-6.

CIS AND TRANS

- **Cis:** the H atoms (and chain parts) on the **same side** of the C=C → the chain **bends**, packs **loosely** → **lower melting point** (olive oil is liquid). Found in **natural** fatty acids (**plant oils, olive oil**): **good** for health.
- **Trans: opposite sides** → straighter chain, larger contact surface, packs **compactly** → **higher melting point** (butter is solid). From **processed food** (french fries, hot dogs, fast food, butter): **bad**, because trans fats **decrease HDL and increase LDL**.
- **Industrial trans fats** are made by **partial hydrogenation** and are now **banned** (labels may say "0% trans fat").

Slide exercise

Physical properties (melting point) of fatty acids depend on **chain length** and **degree of unsaturation**: answer **d, all of the above**.

FATS AND OILS

- **Fat:** a triacylglycerol mixture that is **solid or semi-solid at room temperature (25 °C)**; generally from **animal** sources; **saturated** fatty acids predominate.
- **Oil:** a triacylglycerol mixture that is **liquid at 25 °C**; generally from **plant** sources; larger amounts of **mono- and polyunsaturated** fatty acids.

ESSENTIAL AND NONESSENTIAL FATTY ACIDS

- **Nonessential:** can be **synthesized in the body** (liver).
- **Essential:** must come from the **diet** because the body cannot make them; **precursors of eicosanoids**; needed for proper **membrane structure**; starting materials for longer omega-6 and omega-3 fatty acids.
- The two most important: **linoleic acid (18:2), omega-6** and **linolenic acid (18:3), omega-3**.

SUMMARY TABLE: ESSENTIAL FATTY ACIDS: OMEGA-6 VS OMEGA-3

Family	Dietary essential fatty acid	Dietary sources	Made into	Effect of its eicosanoids
Omega-6	**linoleic acid** (18:2)	meat	**arachidonic acid** (meat, eggs) → eicosanoids	**increase** inflammation and blood pressure
Omega-3	**linolenic acid** (18:3)	soy, walnuts, flaxseed	**eicosapentaenoic acid, EPA** (fish oil) → eicosanoids	**decrease** inflammation and blood pressure

- **Omega-6** eicosanoids **increase** inflammation and blood pressure; **omega-3** eicosanoids **decrease** inflammation and blood pressure.

TRIACYLGLYCEROLS (TRIGLYCERIDES)

- **Glycerol + three fatty acids** joined by **ester bonds**. Glycerol has **three OH groups** (one per fatty acid); a wax's alcohol has **one** OH (one fatty acid).
- Function: **energy storage**. Stored mainly in **adipocytes**; **adipose tissue** is under the skin, in the abdominal cavity, in the mammary glands and around organs. Also an **insulator** (traps heat) and **shock absorber** (protects organs from force).
- The body's fat lets a normal person **survive starvation for 2 or 3 months** (you still lose weight fast as fat is burned by β-oxidation; water is the real essential).

SIMPLE VS MIXED TRIACYLGLYCEROLS

- **Simple triacylglycerol:** a triester of glycerol with **three identical** fatty acids.
- **Mixed triacylglycerol:** glycerol with **more than one kind** of fatty acid (look at the nonpolar tails:

if they differ, it is mixed). "Simple" here is not the same as "simple lipid".

ESTERIFICATION

- **One glycerol + three fatty acids → triacylglycerol + 3 H_2O**: each of the three OH groups is esterified (the glycerol OH links to the fatty acid's carboxyl group). It is a dehydration, but the specific name is **esterification** (it forms the **ester linkage**).

WHAT TO REVIEW

- Definition (insoluble in water, soluble in nonpolar solvents); no common structure, not polymers.
- Five functions with examples (TAG; phospholipids/sphingoglycolipids/cholesterol; bile acids; steroid hormones/eicosanoids; waxes); bile: liver → gallbladder → small intestine.
- Simple vs complex vs derived, with the Figure 19.1 structures and their linkages (ester vs amide).
- Bilayer (polar heads out, tails in); cholesterol and fluidity; sphingosine (18 C, monounsaturated aminodiol); sphingomyelin (myelin), cerebroside (brain); HDL vs LDL.
- Fatty acids: amphipathic, even, unbranched, monocarboxylic; saturated/mono/poly; omega-3 vs omega-6 counting; cis vs trans (shape, melting point, health).
- Fats vs oils; essential fatty acids and their sources; TAG structure, storage, simple vs mixed, esterification.